

Diabetes Prevalance in Virginia

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Introduction

1. Do Virginia counties with higher rates of physical inactivity have higher rates of adult diabetes prevalence?
2. Do Virginia counties with higher poverty levels have correspondingly higher rates of adult diabetes prevalence?
3. Do Virginia counties with a higher percentage of college-educated adults have a lower prevalence of Type 2 diabetes?

Diabetes is one of the most pressing public health challenges facing Virginia today. In 2023, the prevalence of diabetes among Virginia adults stood at 11.8%, following a steady increase since 2016. Each year, an estimated 51,233 Virginians are newly diagnosed, and the disease generates billions in direct medical costs alongside an additional \$2.3 billion in lost productivity. Because this burden is not distributed evenly across the Commonwealth, identifying the county level factors that affect prevalence is critical, which is why we chose this as our topic. This analysis investigates three specific questions. First, whether counties with higher physical inactivity rates have higher diabetes prevalence, given that inactivity is one of the most well-established modifiable risk factors for Type 2 diabetes. Second, whether higher county level poverty corresponds to higher diabetes rates, as adults with incomes below \$25,000 experience diabetes at more than twice the rate of those earning \$75,000 or more. Third, do counties with more college educated adults have lower diabetes prevalence, since Virginia adults without a high school diploma have a diabetes prevalence of 14.6%, nearly double the 8.1% seen among college graduates. Together, these questions target the behavioral and socioeconomic conditions most likely to explain why diabetes is far more prevalent in some parts of Virginia than others.

Data Summary

Metadata Description

The data for this analysis relies on several primary public health and demographic data sources aggregated at the county level for the state of Virginia. Adult diabetes prevalence, physical inactivity rates, and obesity rates were collected from the Centers for Disease Control and Prevention (CDC) PLACES dataset, which utilizes self-reported health surveys. Demographic and socioeconomic variables, including the percentage of the population aged 65 and

older, poverty levels, and educational attainment, were sourced from the National Institutes of Health (NIH) HD Pulse and the University of Wisconsin Population Health Institute's County Health Rankings. Median household income data was compiled using Google Data Commons. Because this data is aggregated at the county level from self-reported surveys and census estimates, there are minor reliability concerns regarding inherent sampling error and reporting bias

EDA Summary

The response variable, diabetes prevalence, is appropriate for multiple linear regression because it is quantitative, continuous, and measured as a percentage. The histogram of diabetes prevalence shows a roughly unimodal distribution with slight right skewness. There are no severe outliers or gaps requiring transformation of the response variable. Alongside our response variable, we collected response variables including `inactivity_rate`, `obesity_rate`, `median_household_income`, `pct_65_over`, `urban_rural`, `health_insurance_coverage`, and `poverty_level`. Among these variables, physical inactivity rate and obesity rate demonstrated the strongest visual relationship with the response variable, diabetes prevalence. The scatterplot of inactivity rate against diabetes prevalence revealed a very strong positive linear relationship (correlation near 0.90), showing that counties with higher inactivity rates have substantially higher diabetes prevalence. During the model-building phase, a squared transformation of inactivity and obesity rate was found to give a linear relationship between the predictors and the response variable, improving correlation and linearity. Median household income showed a moderate negative relationship with diabetes prevalence, but this relationship was not linear; a square root transformation of income improved the linearity and correlation. Other explanatory variables were not needing of any transformations to improve linearity/correlation, or transformations that were tested did not show any improvement. There were also two interaction plots that revealed a possible role in model building: `obesity_rate_sq` and `poverty_level` (visualized in Appendix C), and `health_insurance_coverage` and `poverty_level` (not visualized). We tested these interactions in the later stages of model building.

Analysis and Results

The analysis process utilized multiple linear regression to predict county-level adult diabetes prevalence. Model building was performed iteratively using significance testing, supported by our initial variable screening. First, a full quantitative model was fit, which included squared

transformations for inactivity rate, obesity rate, and percent aged 65 and over, alongside the square root of median household income. A global F-test and subsequent individual t-tests yielded p-values greater than 0.05 for both education and income, indicating they were not statistically significant after accounting for the other predictors. Iterative screening via stepwise regression confirmed these findings by removing variables that failed to meet the $p < 0.05$ significance threshold, resulting in a refined quantitative model retaining only the squared inactivity rate, the squared obesity rate, and the percentage of the population aged 65 and older.

Next, after fitting the quantitative model, variance inflation factors (VIF) were calculated to test for multicollinearity. The results indicated no severe multicollinearity among the predictors, as all individual VIF values fell well below the standard threshold of 10. Specifically, the squared inactivity rate had a VIF of 3.495, the squared obesity rate had a VIF of 3.477, and the percentage of the population aged 65 and over had a VIF of 1.012. Overall, the quantitative model demonstrated a low mean VIF of approximately 2.66.

Next, a separate qualitative model assessed urban/rural classification, poverty level, and health insurance coverage. An individual t-test for urban/rural status yielded a p-value of 0.2878. Because this is greater than the 0.05 significance level, urban/rural status was deemed not statistically significant and was removed. Subsequently, a nested F-test was utilized to evaluate the overall significance of the remaining categorical variables. This test revealed that poverty level was a highly significant predictor, producing a p-value of $8.968e-14$. Since this is strictly less than 0.05, it confirmed that poverty level must be retained. In contrast, the nested F-test for health insurance coverage produced a p-value of 0.1334. Because this value exceeds 0.05, it mathematically demonstrated that health insurance coverage did not significantly improve the overall model fit once poverty level was already accounted for, leading to its removal. Thus, poverty level alone was retained as the sole qualitative predictor for the final model.

The significant quantitative and qualitative predictors were combined to test for interactions. Because exploratory data analysis suggested the effect of obesity might differ across poverty categories, an interaction between obesity rate and poverty level was added. A nested F-test comparing the additive main-effects model to the interaction model produced $F = 8.91$ with a p-value of 0.00024. This is well below the 0.05 significance level, indicating the interaction significantly improved the model fit. Furthermore, the individual t-test for the coefficient of obesity rate in low-poverty counties was statistically significant ($p < 0.05$) and negative, confirming the positive relationship between obesity and diabetes prevalence is weaker in low-poverty counties. An additional interaction between health insurance coverage and poverty

level was tested, but an ANOVA test yielded a p-value of 0.201. Because this p-value is greater than 0.05, we failed to reject the null hypothesis, and this interaction was excluded.

Residual analysis was conducted to check regression assumptions and actively identify extreme or influential observations. To strictly evaluate for extreme observations, leverage (hat values), Cook's distance, and deleted studentized residuals were calculated. Specific cutoffs were established based on our sample size of 133 observations and 8 model parameters: a leverage threshold of 0.120, a Cook's distance threshold of 0.030, and an absolute studentized residual threshold of 2. During this assessment, observations 103, 110, and 125 were identified as highly influential outliers because they simultaneously exceeded all three of these diagnostic metrics. Because these specific counties exerted a disproportionate and undue influence on the model's coefficients, they were formally removed from the dataset.

The impact of removing these outliers produced a demonstrably stronger and more reliable final regression model. By refitting the equation without these highly influential data points, the overall fit of the model improved, with the Adjusted R-squared increasing from 0.9057 to 0.9160. Furthermore, the Residual Standard Error decreased noticeably from 0.5841 to 0.5392, confirming that the model's average prediction errors are smaller and the remaining observations sit much closer to the fitted regression line. Without the disproportionate leverage of those specific outliers, the standard errors for the coefficients tightened, which resulted in smaller p-values for key predictors. Most notably, the base obesity rate variable shifted from failing the standard significance threshold ($p = 0.0628$) to becoming statistically significant ($p = 0.01499$). Finally, the removal of these outliers visually improved the diagnostic plots. In the original residuals versus fitted values plot, a cluster of extreme positive residuals pulled the trend line upward; in the cleaned plot, the removal of those specific points flattens the trend line, demonstrating a more random, stable scatter that better satisfies the assumption of constant variance. Similarly, the refitted model's normal Q-Q plot exhibits a much tighter and more consistent alignment with the theoretical diagonal line compared to the original, confirming that the normality assumption is now robustly satisfied for accurate hypothesis testing.

To fulfill the additional technique requirement, External Model Validation was performed by evaluating the Predicted Residual Error Sum of Squares (PRESS) statistic. The PRESS statistic evaluates the model's predictive ability by calculating the sum of squared prediction errors. The additive model produced a PRESS value of 50.53, while the chosen interaction model produced a lower PRESS value of 49.95, confirming superior predictive performance on hold-out data.

We ultimately selected $\text{lm}(\text{diabetes_prevalence} \sim \text{obesity_rate_sq} * \text{poverty_level} + \text{obe-}$

sity_rate * poverty_level + inactivity_rate_sq + inactivity_rate + pct_65_over, data = df) as our final model because it achieved a strong Adjusted R-squared of 0.916 (explaining roughly 91.6% of the variance in diabetes prevalence), an excellent Residual Standard Error of 0.5392, and an optimized PRESS statistic of 49.95.

Conclusions

Our final prediction equation is: $\hat{Y} = 27.194 + 0.017(\text{obesity_rate}^2) - 16.517(D_{Low}) + 1.003(D_{Med}) - 1.183(\text{obesity_rate}) + 0.004(\text{inactivity_rate}^2) + 0.116(\text{inactivity_rate}) - 0.040(\text{pct_65_over}) - 0.011(\text{obesity_rate}^2 \cdot D_{Low}) + 0.001(\text{obesity_rate}^2 \cdot D_{Med}) + 0.863(\text{obesity_rate} \cdot D_{Low}) - 0.082(\text{obesity_rate} \cdot D_{Med})$, where $D_{Low} = 1$ if Low Poverty and $D_{Med} = 1$ if Medium Poverty (High Poverty is baseline).

The obesity rate squared coefficient of 0.017 indicates a nonlinear, accelerating relationship. As obesity increases, each additional unit contributes more to diabetes prevalence than the previous, confirming obesity as a dominant driver of adult diabetes prevalence. The pct_65_over coefficient of -0.040 indicates that for each one percentage point increase in the population aged 65 and over, estimated diabetes prevalence decreases by 0.040 percentage points, likely reflecting demographic and survivorship patterns at the county level. The model explains 91.6% of variation in diabetes prevalence (Adjusted $R^2 = 0.916$) and is highly significant ($p < 2.2 \times 10^{-16}$), with a residual standard error of 0.539 percentage points, which is a useful precision for county-level public health planning. As a prediction example, a High Poverty county with 38% obesity rate, 28% inactivity rate, and 14% population aged 65 and over is predicted to have approximately 12.6% diabetes prevalence.

Returning to our research questions: counties with higher obesity rates do have significantly higher diabetes prevalence, confirming our first question. Poverty level was retained in the model but was not individually significant after controlling for obesity, suggesting its effect is largely mediated through obesity rather than acting independently, partially answering our second question. Education was removed during step wise selection, indicating it does not independently predict diabetes prevalence after accounting for other variables. This analysis is limited by its cross-sectional, county-level, one-year focused design, restricting generalization. Better variables such as food access, healthcare density, and smoking rates were unavailable for the same year. Future work should expand to all U.S. states, incorporate longitudinal data, and include additional lifestyle variables to better explain remaining variation in diabetes prevalence.

Appendix A: Data Dictionary

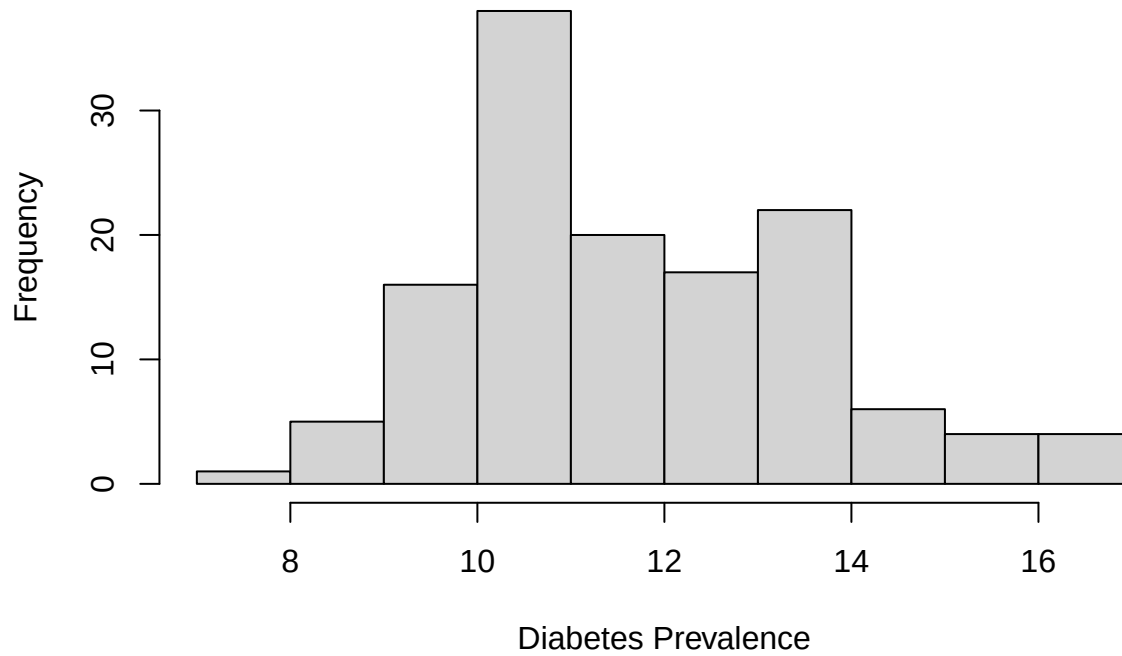
Variable Name	Abbreviated Name	Description
Adult Diabetes Prevalence	diabetes_prevalence	Percentage of adults in the county diagnosed with diabetes. This is the response variable.
Inactivity Rate	inactivity_rate	Percentage of adults who report no leisure-time physical activity.
Median Household Income	median_household_income	The median yearly household income in the county, measured in dollars.
Population 65 and Over	pct_65_over	Percentage of the county population aged 65 or older.
Obesity Rate	obesity_rate	Percentage of adults classified as obese (BMI of 30 or higher).
Less Than HS Education	pct_less_than_hs	Percentage of adults aged 25 and older with less than a high school diploma.
Urban/Rural Status	urban_rural	Urban (less than 50% rural residents, 54 counties) or Rural (50% or more rural residents, 79 counties).
Health Insurance Coverage	health_insurance_coverage	High: 0-7% uninsured (52 counties); Medium: 7-9% uninsured (52 counties); Low: greater than 9% uninsured (29 counties).
Poverty Level	poverty_level	Low Poverty: 0-9.5% (44 counties); Medium Poverty: 9.5-15.1% (45 counties); High Poverty: greater than 15.1% (44 counties).

Appendix B: Data Rows

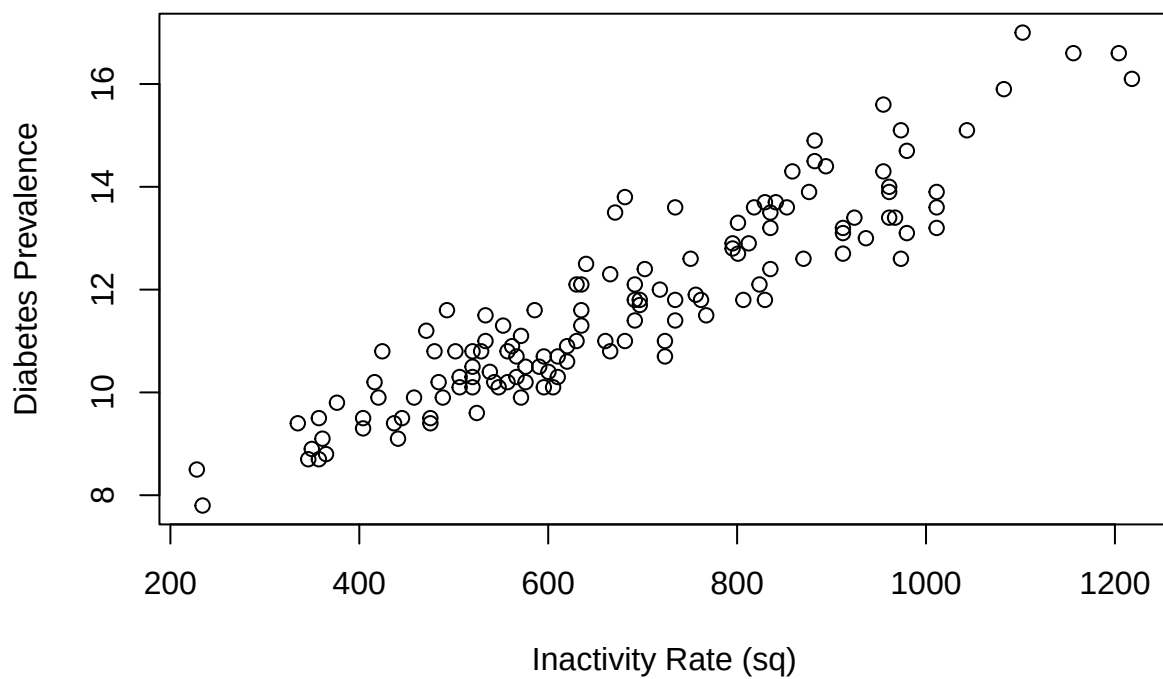
	FIPS	State	County	median_household_income	obesity_rate	pct_65_over
1	51001	Virginia	Accomack	56039	40.3	25.2
2	51003	Virginia	Albemarle	102750	29.0	19.7
3	51005	Virginia	Alleghany	58215	39.1	25.8
4	51007	Virginia	Amelia	71112	38.8	20.5
5	51009	Virginia	Amherst	61499	39.6	21.7
6	51011	Virginia	Appomattox	64701	38.8	20.8
			inactivity_rate	pct_less_than_hs	uninsured_pct	health_insurance_coverage
1			31.0	17.2	12	Low
2			18.6	5.6	7	High
3			25.7	9.9	8	Medium
4			28.2	14.4	10	Low
5			27.1	11.4	7	High
6			26.4	8.0	9	Medium
	poverty_pct	poverty_level	rural_pct	urban_rural	diabetes_prevalence	
1	14.9	Medium Poverty	100.0	Rural	13.4	
2	6.8	Low Poverty	45.0	Urban	8.7	
3	13.1	Medium Poverty	52.4	Rural	11.0	
4	11.1	Medium Poverty	100.0	Rural	12.8	
5	10.8	Medium Poverty	63.7	Rural	11.8	
6	12.3	Medium Poverty	100.0	Rural	11.7	

Appendix C: Exploratory Data Analysis

Histogram of Diabetes Prevalence



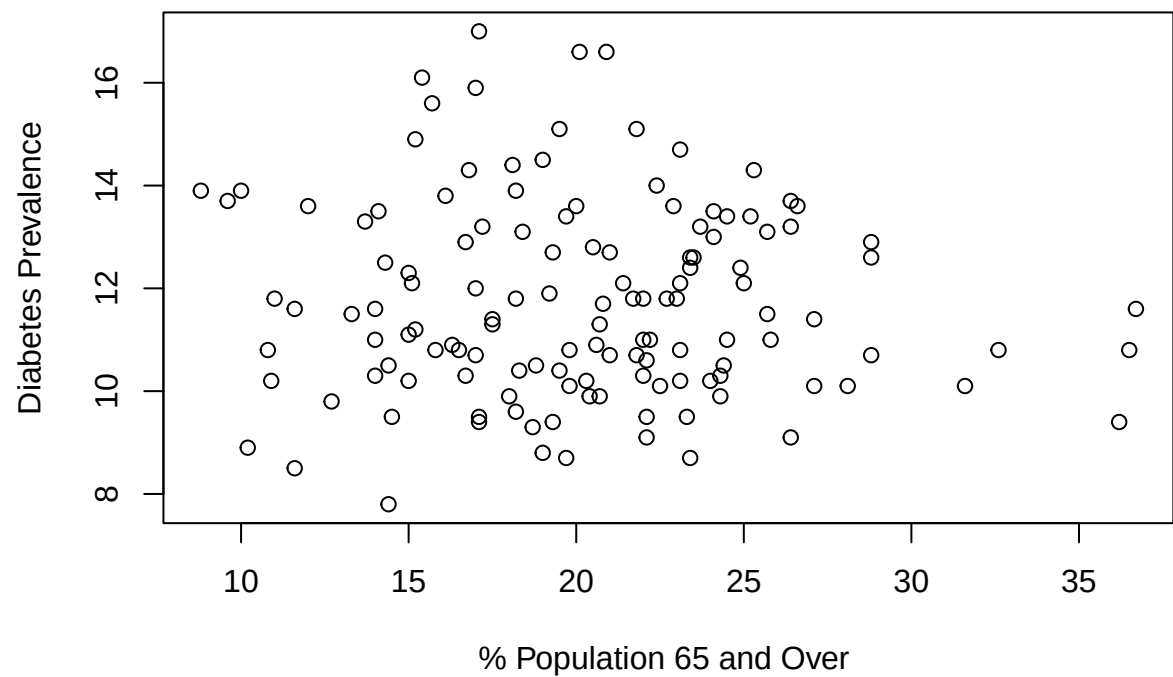
Diabetes Prevalence vs Inactivity Rate (sq)



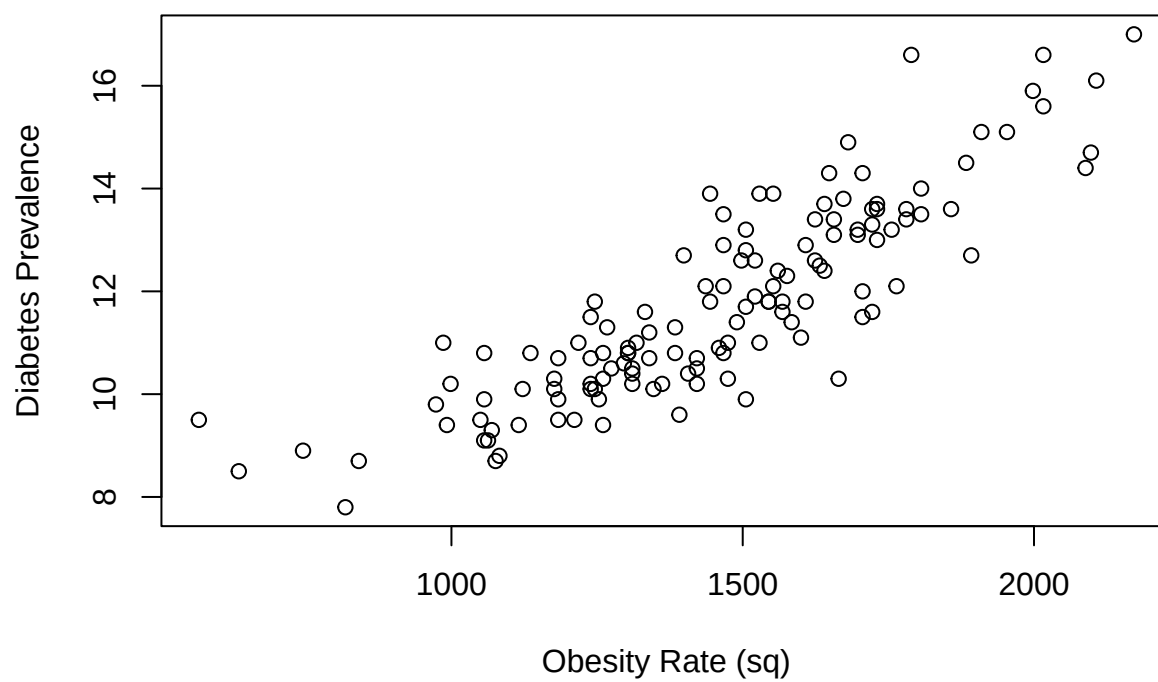
Diabetes Prevalence vs Median Household Income (sqrt)



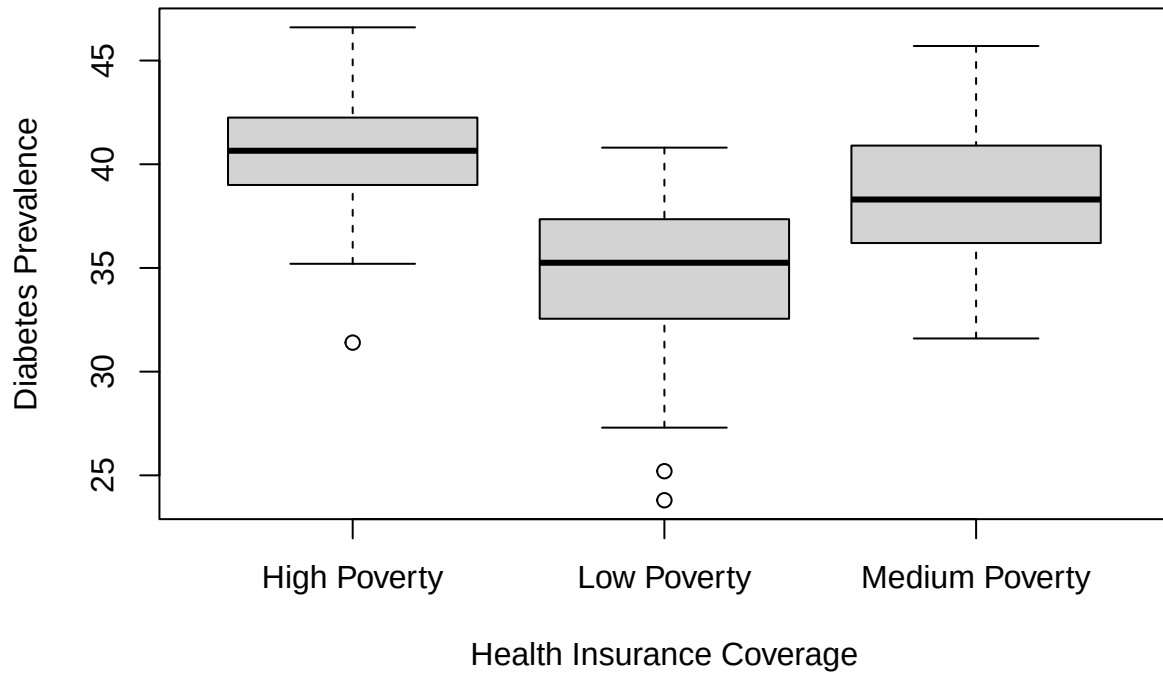
Diabetes Prevalence vs % Population 65 and Over



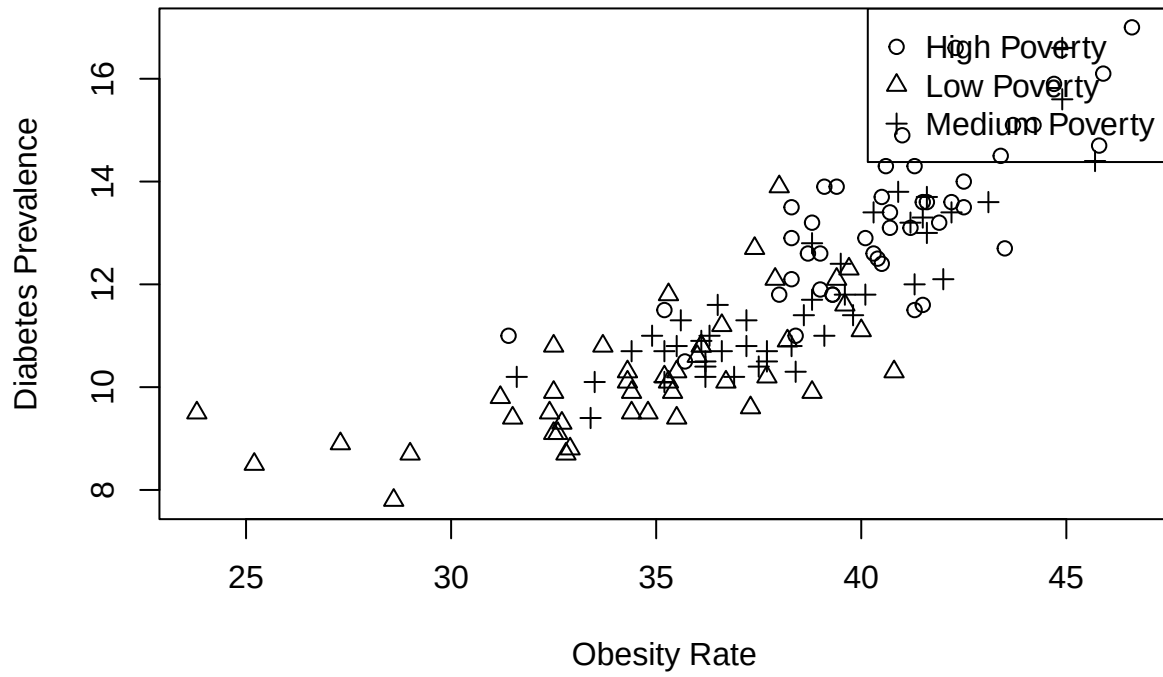
Diabetes Prevalence vs Obesity Rate (sq)



Diabetes Prevalence by Poverty Level

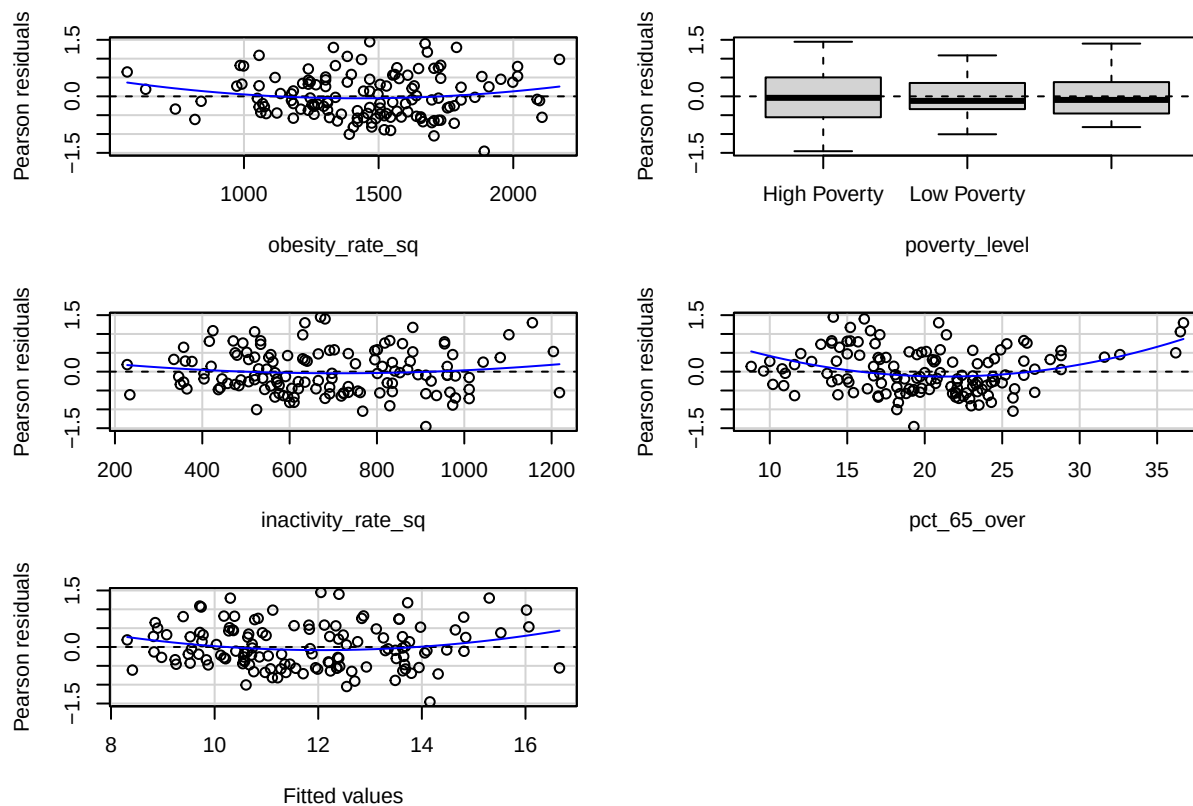


Diabetes Prevalence vs Obesity Rate by Poverty Level

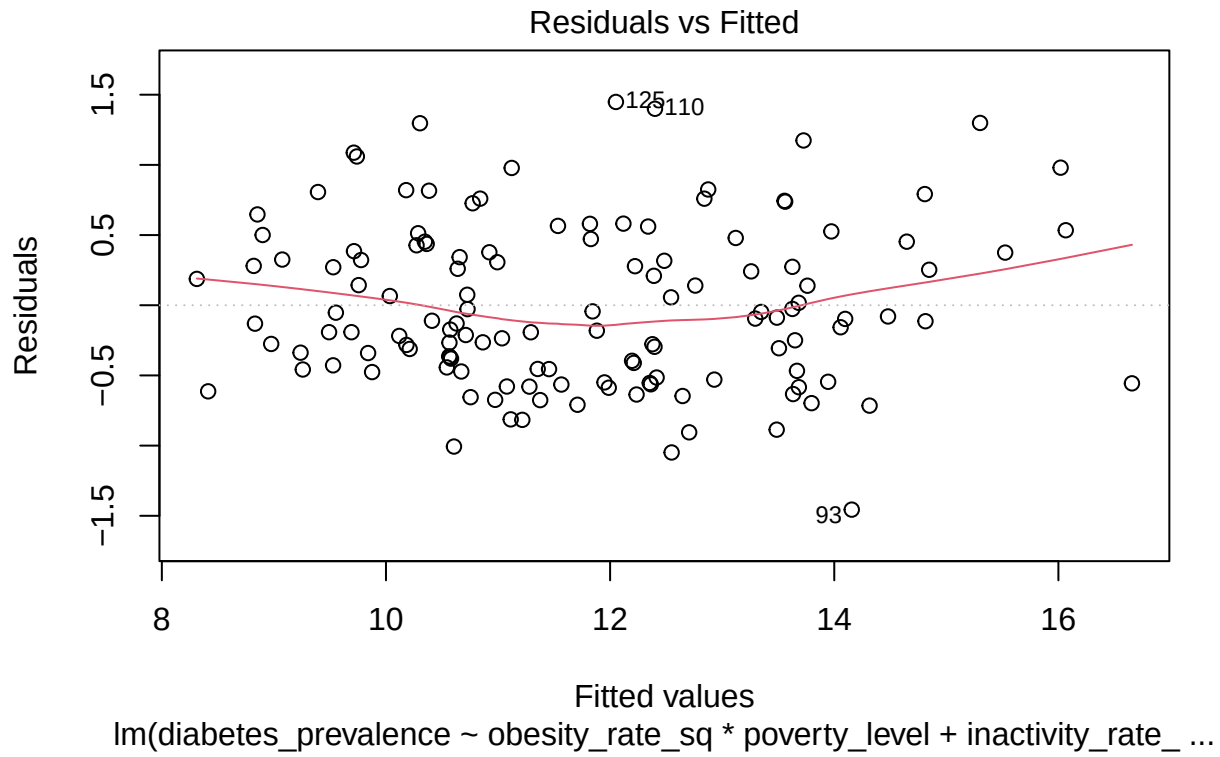


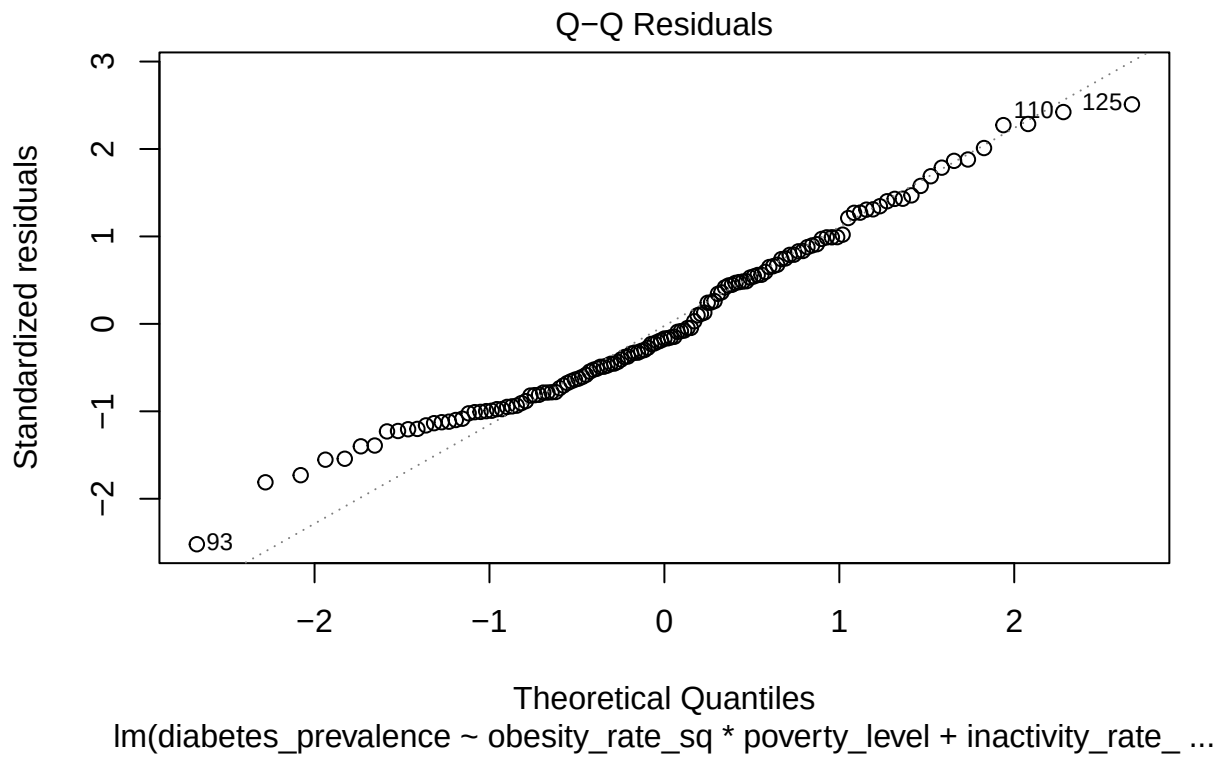
Appendix D: Final Model Output and Plots

```
final_model <- lm(diabetes_prevalence ~ obesity_rate_sq * poverty_level +  
                  inactivity_rate_sq + pct_65_over,  
                  data = df)  
  
summary(final_model)  
library(car)  
library(olsrr)  
  
residualPlots(final_model, tests = F)
```



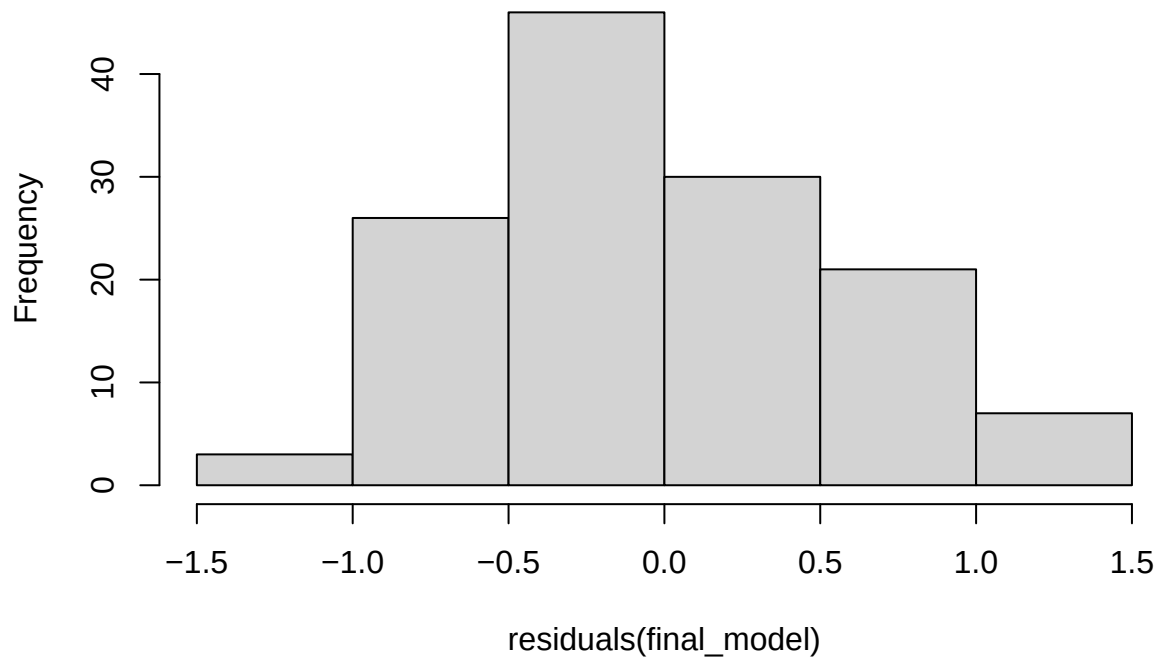
```
plot(final_model, which = c(1,2))
```





```
hist(residuals(final_model))
```

Histogram of residuals(final_model)



```
n <- nrow(df)
p <- length(coef(final_model))

# Leverage threshold = 2p/n
which(hatvalues(final_model) > 2 * p / n)

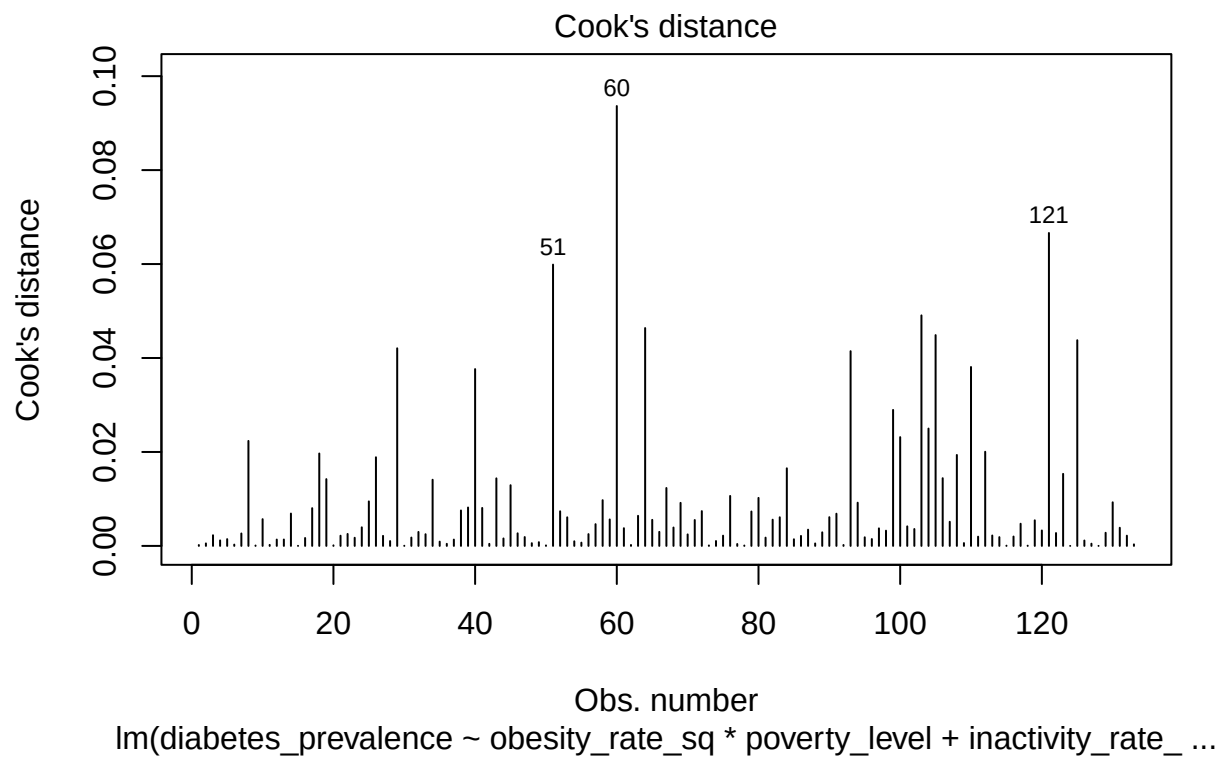
# Cook's Distance threshold = 4/n
which(cooks.distance(final_model) > 4 / n)

# Deleted Studentized Residuals threshold = +/-2
which(abs(rstudent(final_model)) > 2)

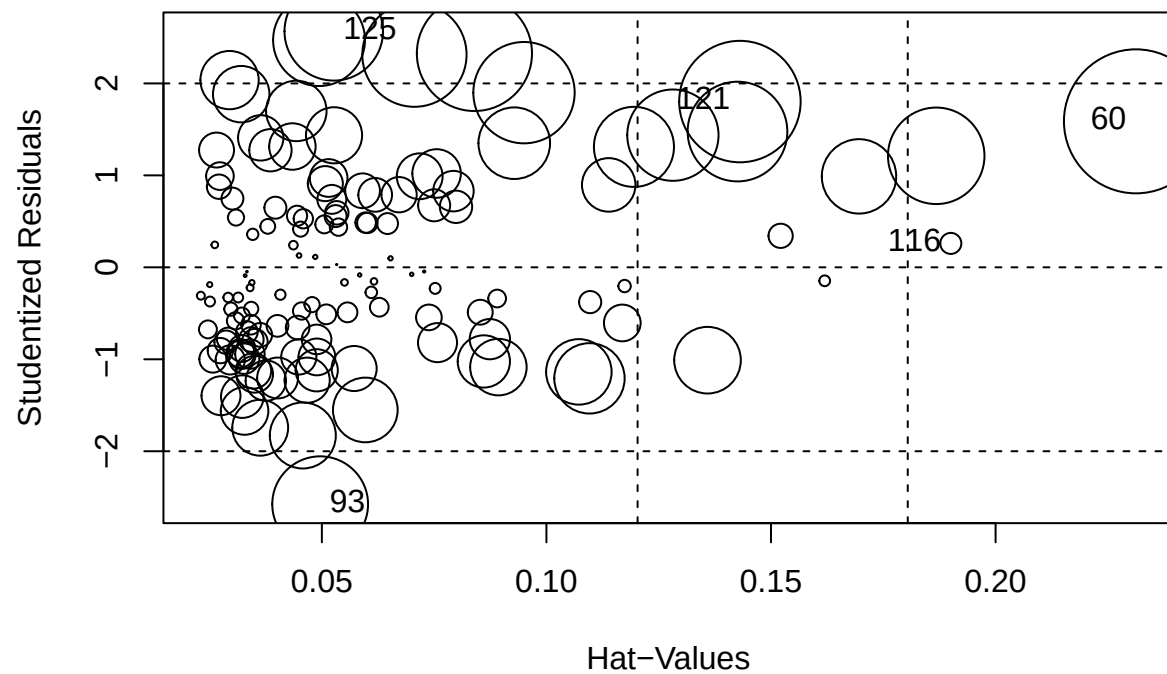
# Observations exceeding ALL three simultaneously
which(
  hatvalues(final_model) > 2 * p / n &
  cooks.distance(final_model) > 4 / n &
  abs(rstudent(final_model)) > 2
```

```
)
```

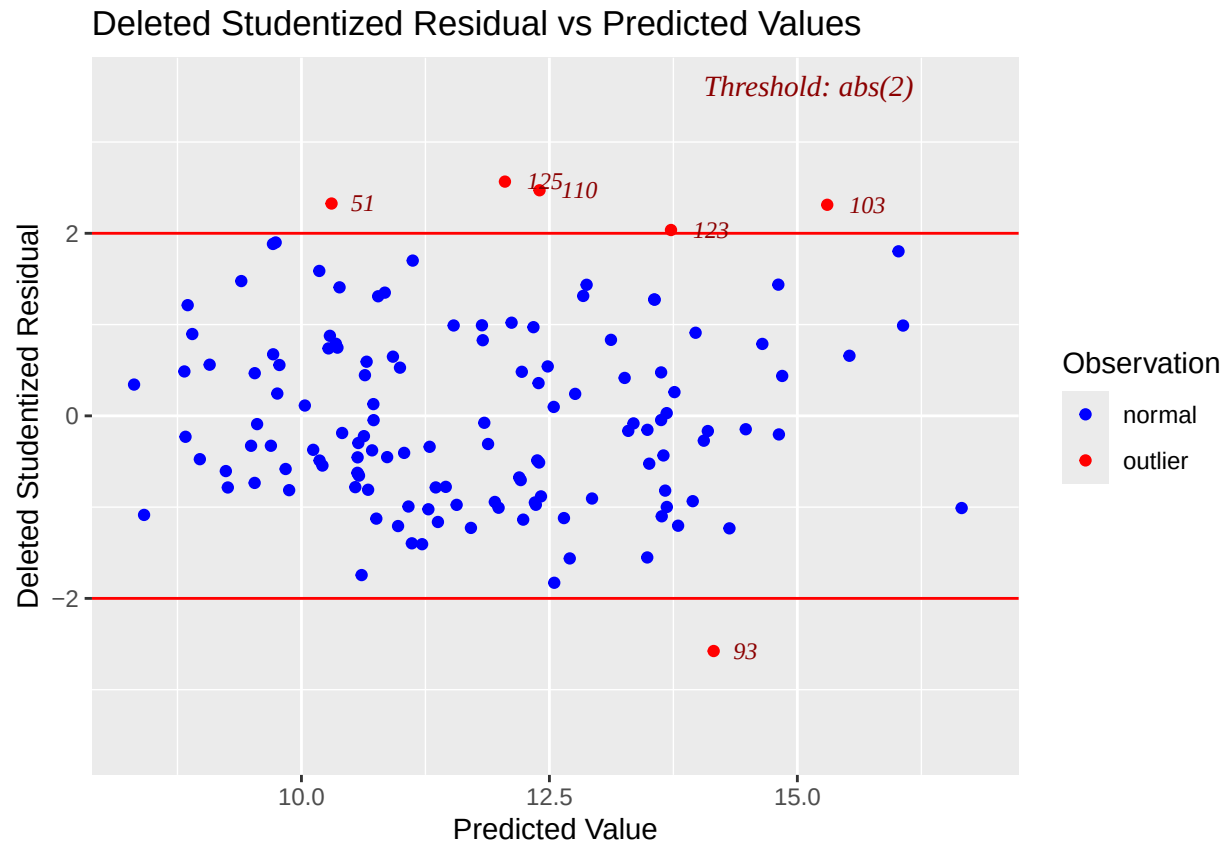
```
plot(final_model, which=4)
```



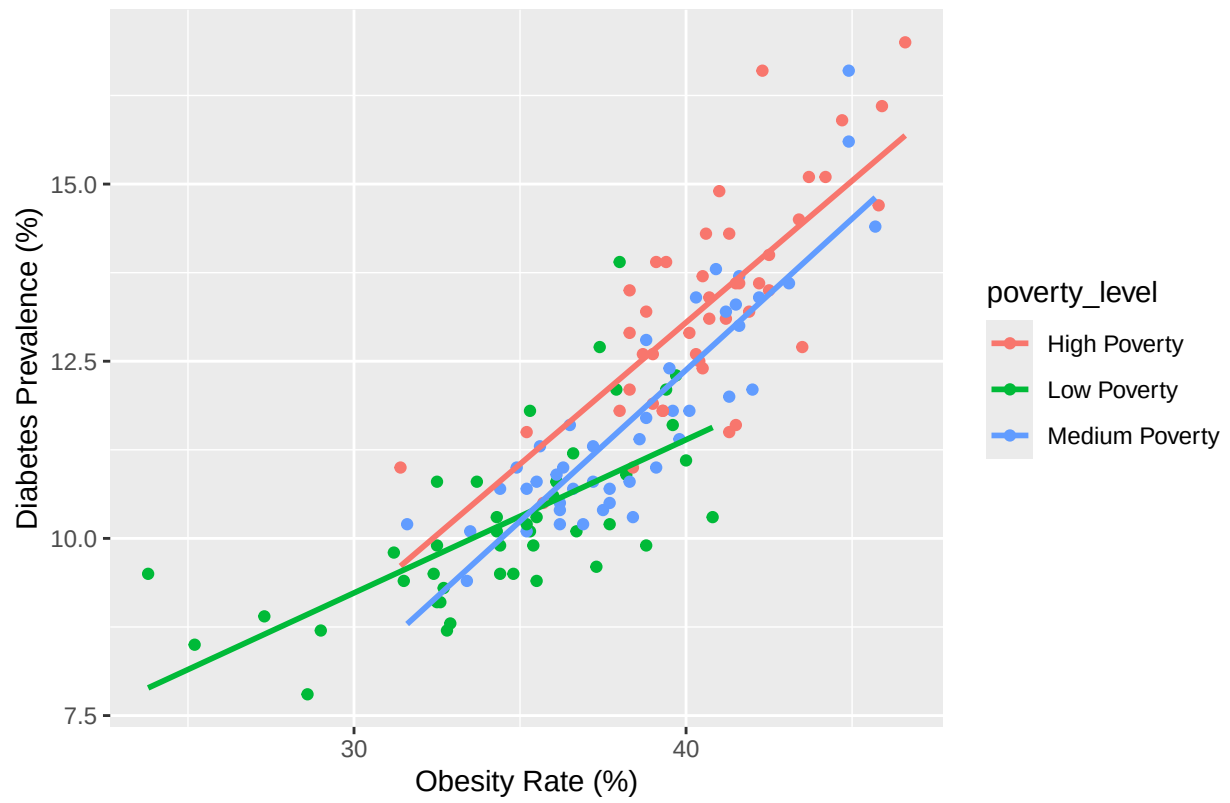
```
influencePlot(final_model, fill=F)
```

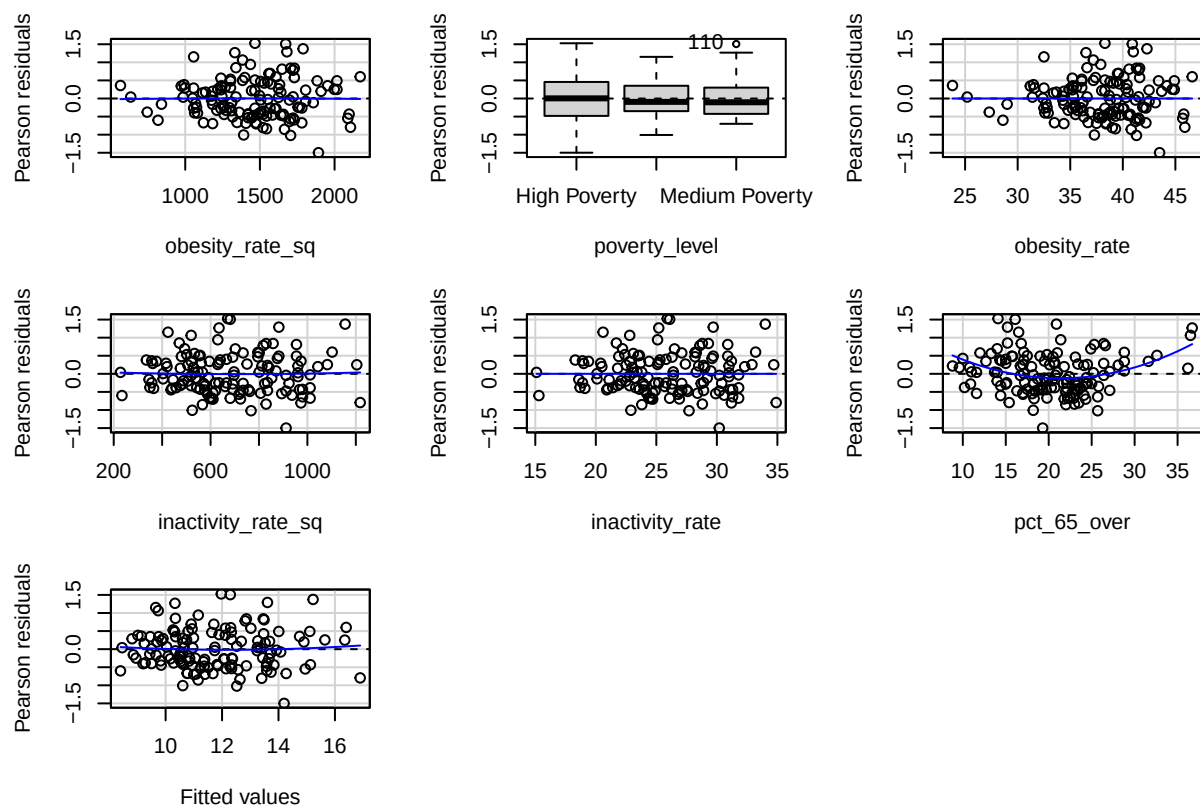


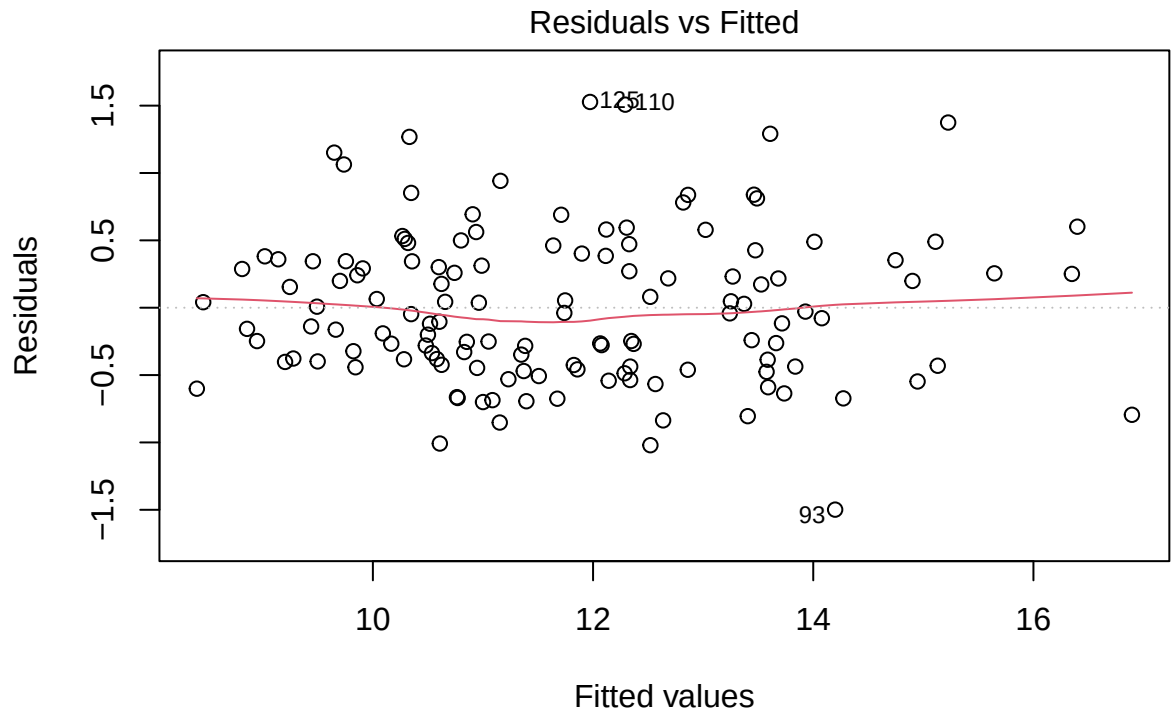
```
ols_plot_resid_stud_fit(final_model)
```



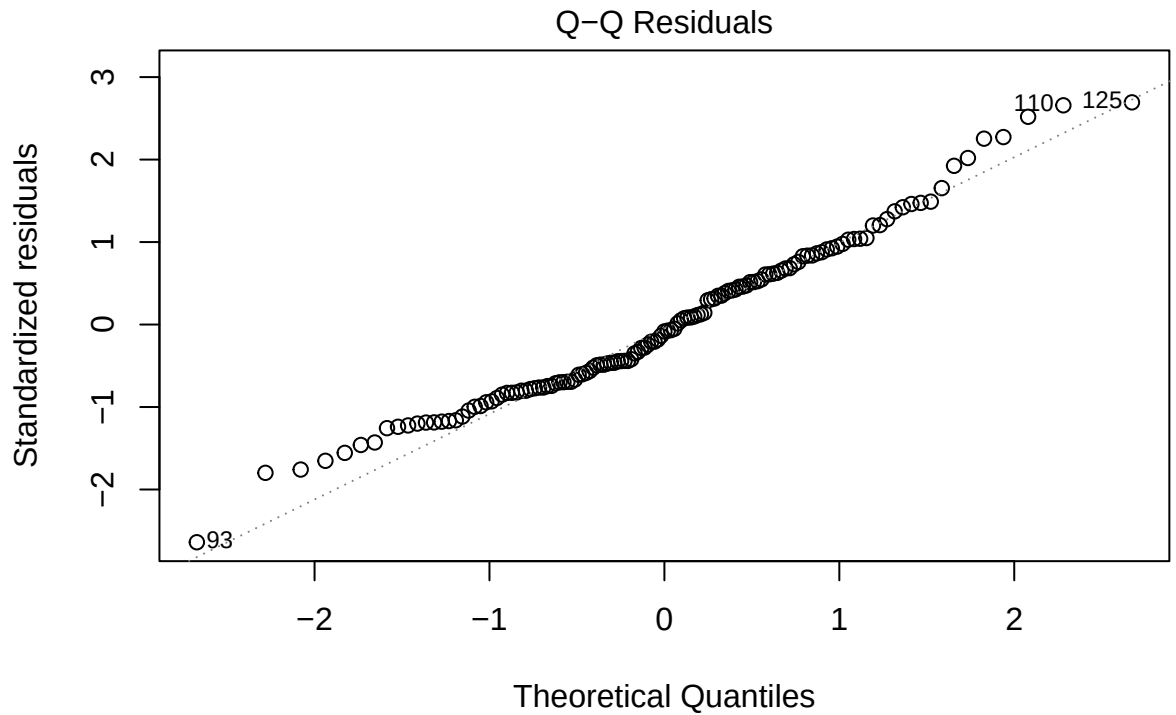
Interaction Between Obesity Rate and Poverty Level





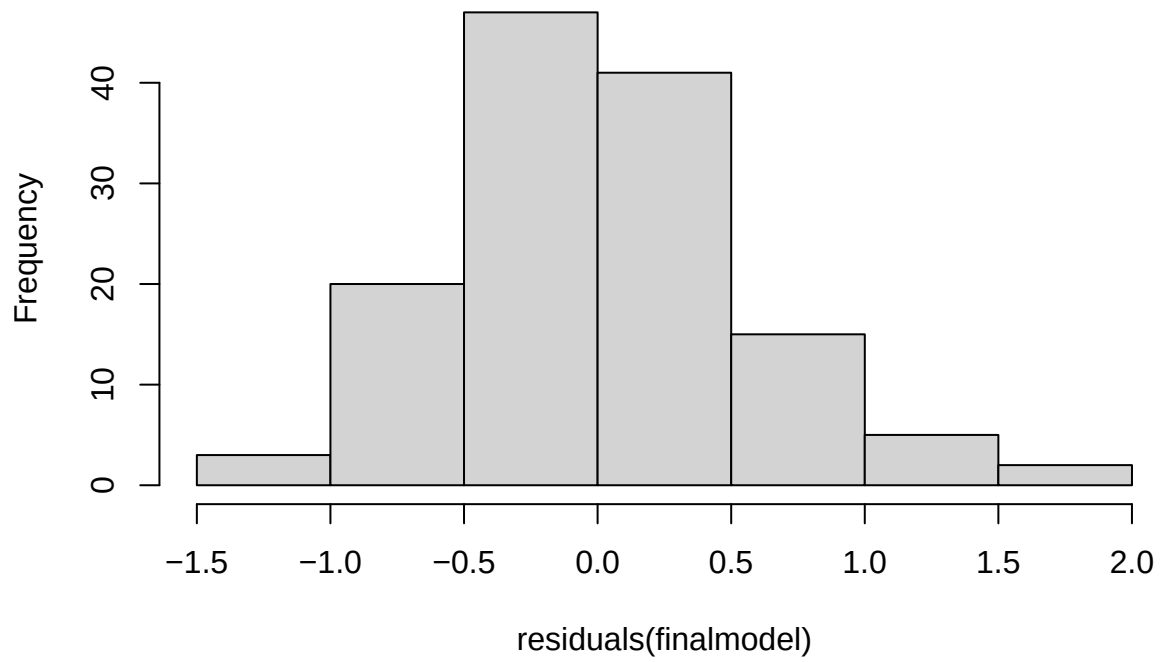


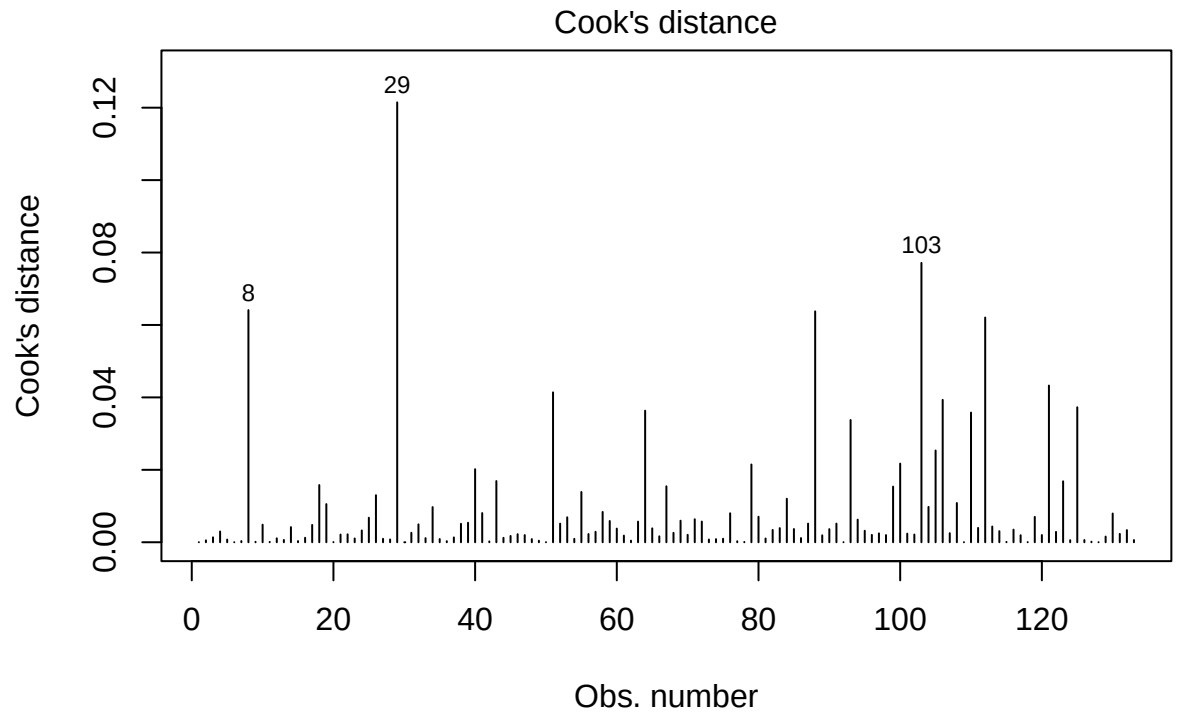
$\text{lm}(\text{diabetes_prevalence} \sim \text{obesity_rate_sq} * \text{poverty_level} + \text{obesity_rate} * \text{p} \dots)$



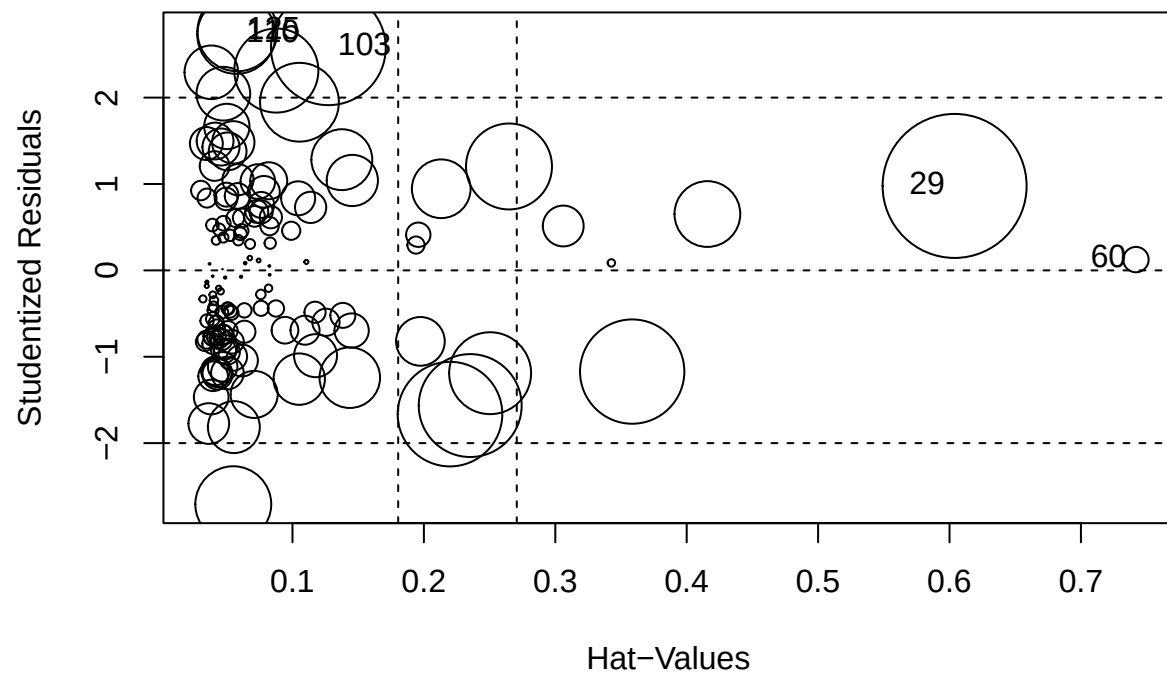
lm(diabetes_prevalence ~ obesity_rate_sq * poverty_level + obesity_rate * p ...

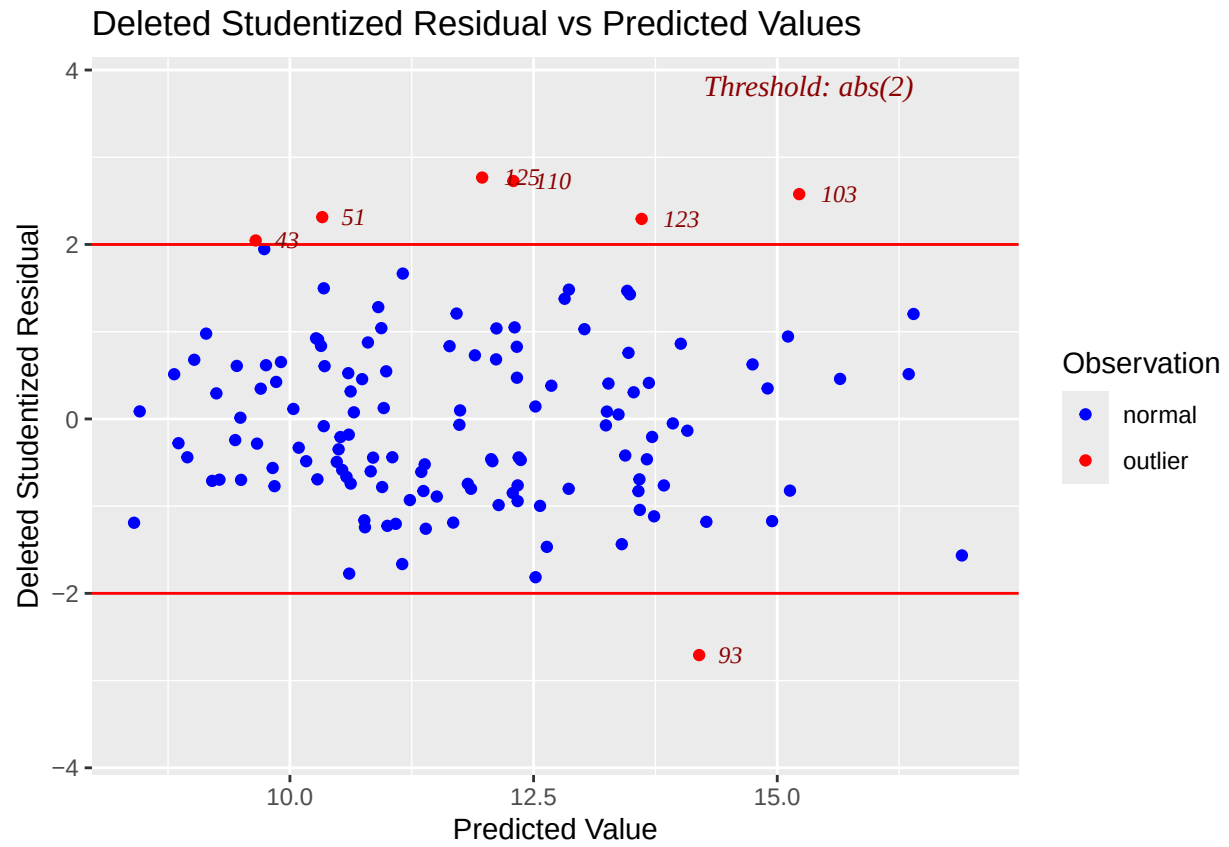
Histogram of residuals(finalmodel)

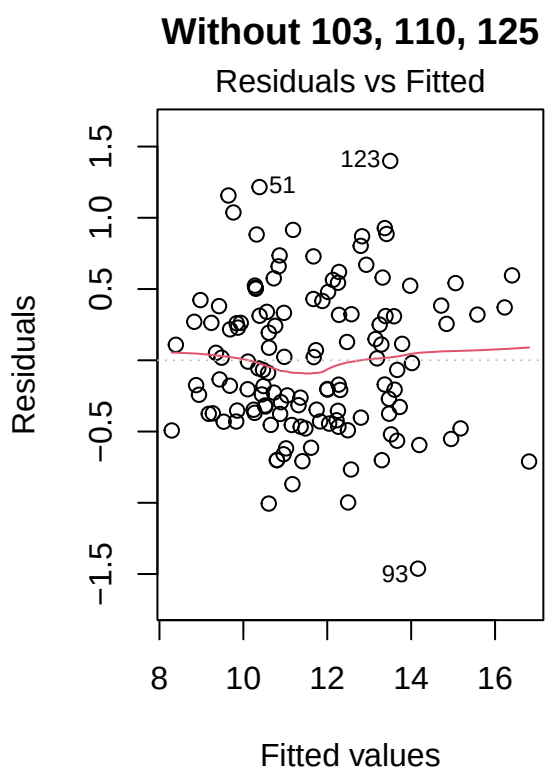
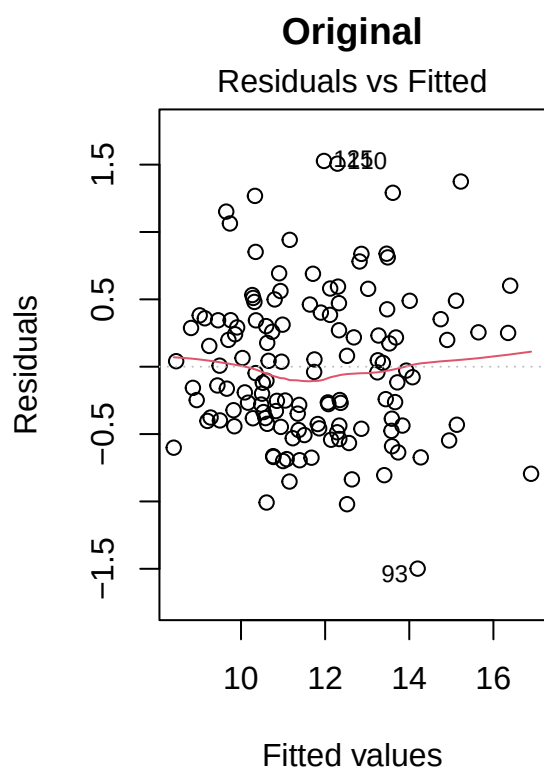


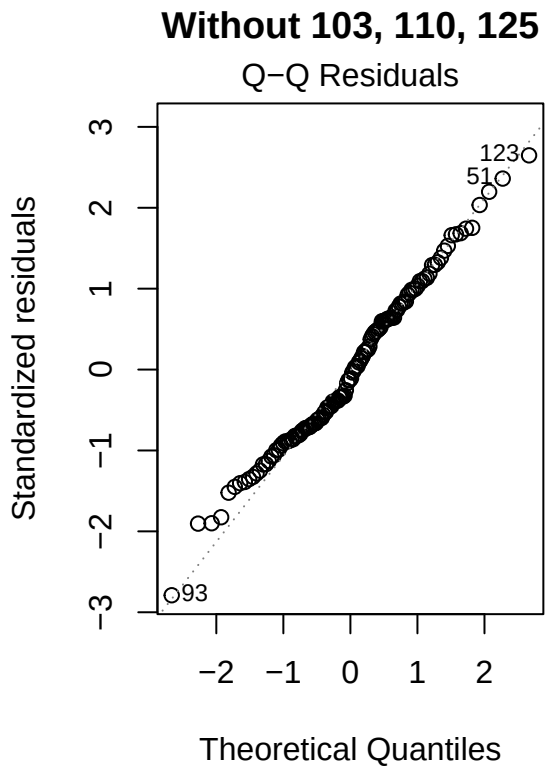
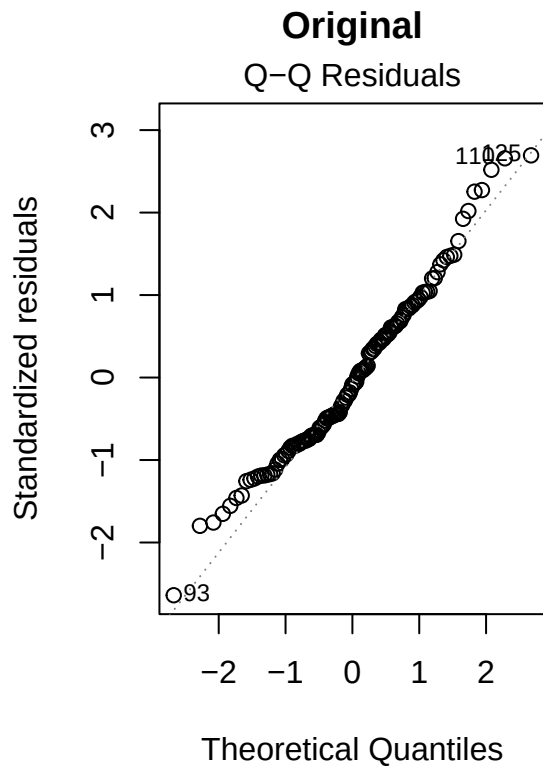


$\text{lm}(\text{diabetes_prevalence} \sim \text{obesity_rate_sq} * \text{poverty_level} + \text{obesity_rate} * \text{p} \dots)$









Appendix E: References

Background

America’s Health Rankings. (2023). Explore diabetes in Virginia. United Health Foundation. <https://www.americashealthrankings.org/explore/measures/Diabetes/VA>

American Diabetes Association. (2023). The burden of diabetes in Virginia. https://diabetes.org/sites/default/files/2023-09/ADV_2023_State_Fact_sheets_all_rev_Virginia.pdf

Centers for Disease Control and Prevention. (n.d.). PLACES: Local data for better health — County data 2023. U.S. Department of Health and Human Services. <https://data.cdc.gov/500-Cities-Places/PLACES-Local-Data-for-Better-Health-County-Data-20/swc5-untb>

Virginia Department of Health. (2020). Data & resources: Diabetes and prediabetes. <https://www.vdh.virginia.gov/diabetes/data/>

Virginia Department of Health. (2023). Chronic disease data dashboard. <https://www.vdh.virginia.gov/data/chronic-disease-in-virginia/>

Data Sources

Centers for Disease Control and Prevention. (n.d.). PLACES: County view. U.S. Department of Health and Human Services. <https://places.cdc.gov/?view=county>

Data Commons. (n.d.). Median household income — Virginia. Google. <https://datacommons.org/explore/#q=median%20income%20household%20virginia>

National Institutes of Health, National Institute on Minority Health and Health Disparities. (n.d.). HD Pulse: Poverty data — Virginia counties. U.S. Department of Health and Human Services. https://hdpulse.nimhd.nih.gov/data-portal/social/table?age=001&age_options=ageall_1&demo=00008&demo_options=poverty_3&race=00&race_options=race_7&sex=0&sex_options=sexboth_1&socialtopic=080&socialtopic_options=social_6&statefips=51&statefips_options=area_states

National Institutes of Health, National Institute on Minority Health and Health Disparities. (n.d.). HD Pulse: Educational attainment data — Virginia counties. U.S. Department of Health and Human Services. https://hdpulse.nimhd.nih.gov/data-portal/social/table?age=081&age_options=age25_1&demo=00005&demo_options=education_3&race=00&race_options=race_7&sex=0&sex_options=sexboth_1&socialtopic=020&socialtopic_options=social_6&statefips=51&statefips_options=area_states

University of Wisconsin Population Health Institute. (2023). County health rankings: Adults 65 and older — Virginia. County Health Rankings & Roadmaps. <https://www.countyhealthrankings.org/health-data/community-conditions/demographics/demographics/65-and-older?state=51&year=2023>

University of Wisconsin Population Health Institute. (2023). County health rankings: Rural population — Virginia. County Health Rankings & Roadmaps. <https://www.countyhealthrankings.org/health-data/community-conditions/demographics/demographics/rural?state=51&year=2023>

Additional Help

Ritz, C., & Spiess, A.-N. (2014). PRESS: PRESS statistic (qpcR version 1.4-2). RDocumentation. <https://www.rdocumentation.org/packages/qpcR/versions/1.4-2/topics/PRESS>